

Experimental units and independence

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Baselines and fair comparisons

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

A literature comparison matrix

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Finding primary research

Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Experimental units and independence. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Experimental units and independence. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Five classrooms each contribute 20 student responses under a classroom-level assignment. Identify assignment units and measurements.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Three queries each run four times produce 12 outputs. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Independence follows design and relationships, not row count. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Five classrooms each contribute 20 student responses under a classroom-level assignment. Identify assignment units and measurements.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled experimental units and independence evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of experimental units and independence. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Experimental units and independence

Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants.

Plant IDs P1–P4 each have three views. All views of a plant must stay in one split.

Keep all views of a plant together

	Plant with 3 views	Plant with 3 views
Train: 6 photos	P1	P2
Evaluate: 6 photos	P3	P4

Read a cell using both its row and column labels.

Experimental units and independence

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the unit

Four plants, not twelve independent plants, generated the observations.



2. Group before splitting

Assign whole plant IDs to training or evaluation.



3. Trace a valid split

Train on P1,P2 and evaluate on P3,P4: six photos per split.

CHANGE ONE THING

Put one P1 view into evaluation while other P1 views train.

Explain the changed result using the matching lesson.

Count independent plants and photo views. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Photos are not always independent plants. A dataset has twelve photos: three views of each of four plants. The intended use is identifying new plants.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Train on P1,P2 and evaluate on P3,P4: six photos per split.

1. Group before splitting
2. Identify the unit
3. Trace a valid split

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Trace a valid split Train on P1,P2 and evaluate on P3,P4: six photos per split. Repeated views from one object share information. Counting them as independent units or scattering them across splits can exaggerate evidence for new-object performance.

Practice 2: explain the deciding evidence

Changed-case practice: Put one P1 view into evaluation while other P1 views train. Which conclusion is supported by the supplied facts?

1. Flag object-level leakage for this new-plant target.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Flag object-level leakage for this new-plant target. Evaluation now includes a plant already represented during training.

Repair a missing building block

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the mechanism and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Review A literature comparison matrix (ai-7/6). Goal: Explain the mechanism and terminology of a literature comparison matrix. Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

Review Finding primary research (ai-7/3). Goal: Explain the mechanism and terminology of finding primary research. Short practice: Separate what the blog claims from what the original report records. Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

